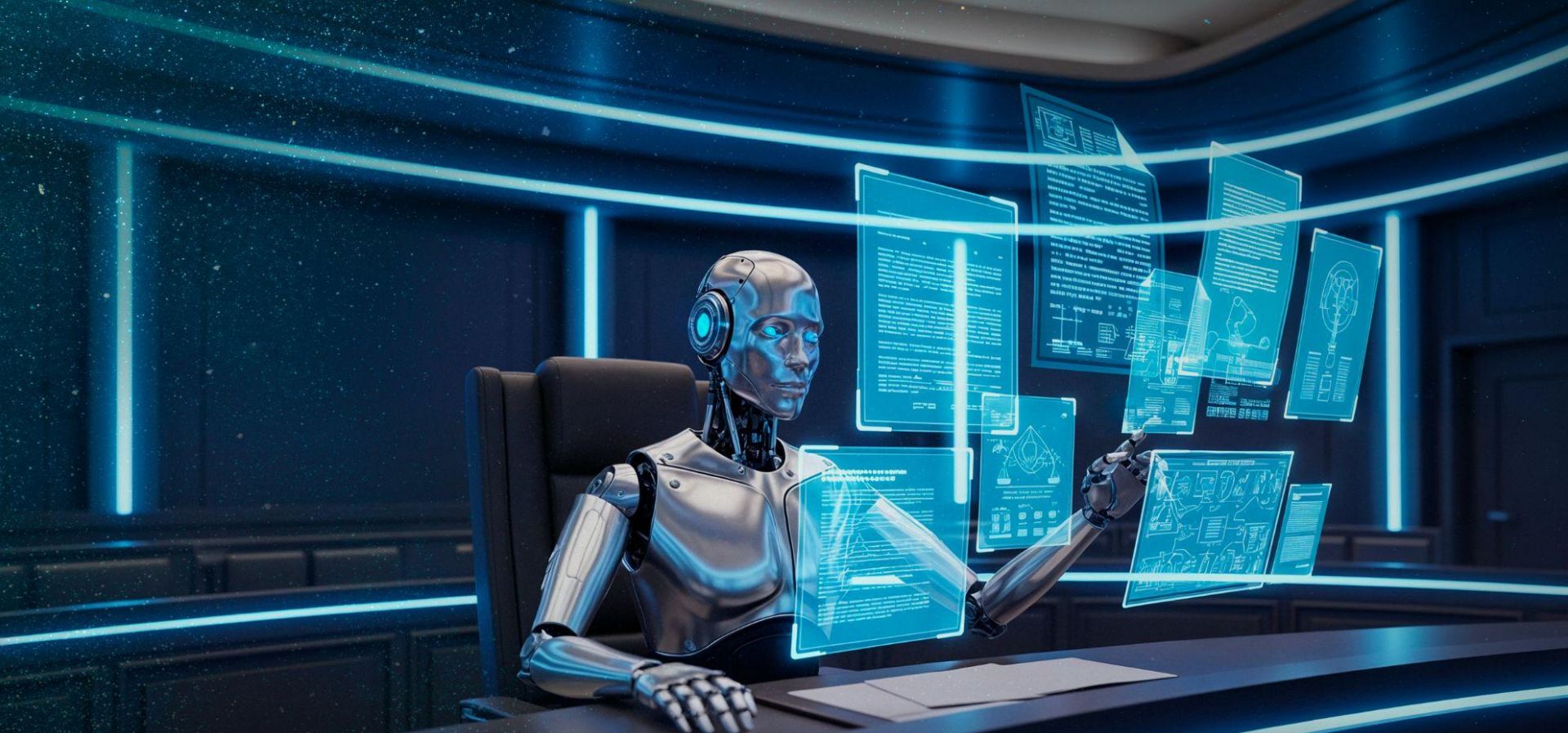




Cybersecurity Frameworks

NIST Cybersecurity Framework (NIST CSF)

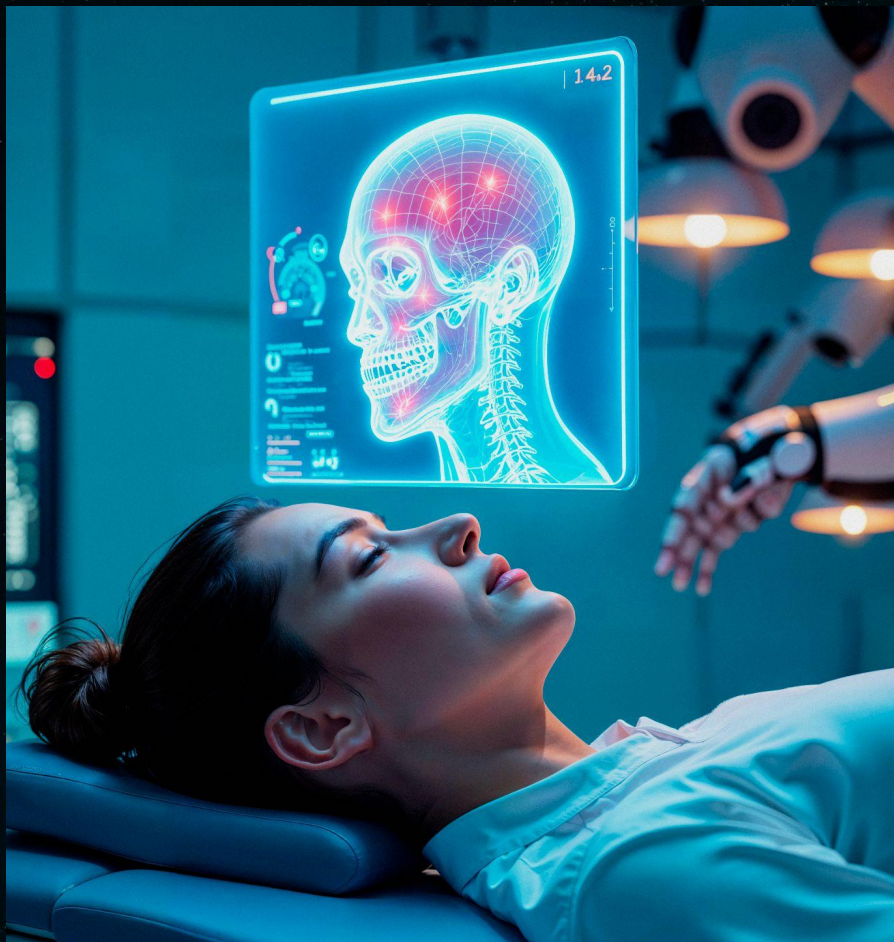
What is NIST?

NIST stands for the National Institute of Standards and Technology.

Think of it as a cybersecurity rulebook created to help organizations protect their systems and data.

NIST breaks cybersecurity into five core functions:

- *Identify – Know what systems, data, and assets you have*
- *Protect – Put controls in place to secure them*
- *Detect – Monitor for security issues*
- *Respond – Take action when an attack happens*
- *Recover – Restore normal operations*



ELITE STAR MEDICAL CLINIC

A small healthcare provider with 50 employees.

- **Stores patient health records digitally**
- **Uses cloud-based applications**
- **Must protect sensitive patient data from cyberattacks**

IDENTIFY: DIGITAL ASSETS



Patient
database:
1,200 records



15
employee
laptops



Google
Workspace
(Google
Drive)



3 admin
accounts
with full
access

EHR
(Electronic
Health
Record)
system

PROTECT

Protection Controls:

- Enable Multi-Factor Authentication (MFA) on all accounts
- Encrypt all patient data at rest and in transit
- Conduct phishing-awareness training every 6 months
- Restrict admin access to only 2 authorized users



DETECT

Detection Methods:

- Install antivirus/EDR on all laptops
- Enable login alerts for after-hours account access
- Use a SIEM tool (e.g., Splunk) to detect unusual network activity
- Review system logs weekly



RESPOND

01

STEP 1

Disconnect infected machines from the network immediately

02

STEP 2:

Notify IT manager and clinic director within 1 hour.

03

STEP 3

Contact cybersecurity incident response team.

04

STEP 4

Do not pay ransom.

05

STEP 5

Preserve logs as evidence.

RESPONSE PLAN



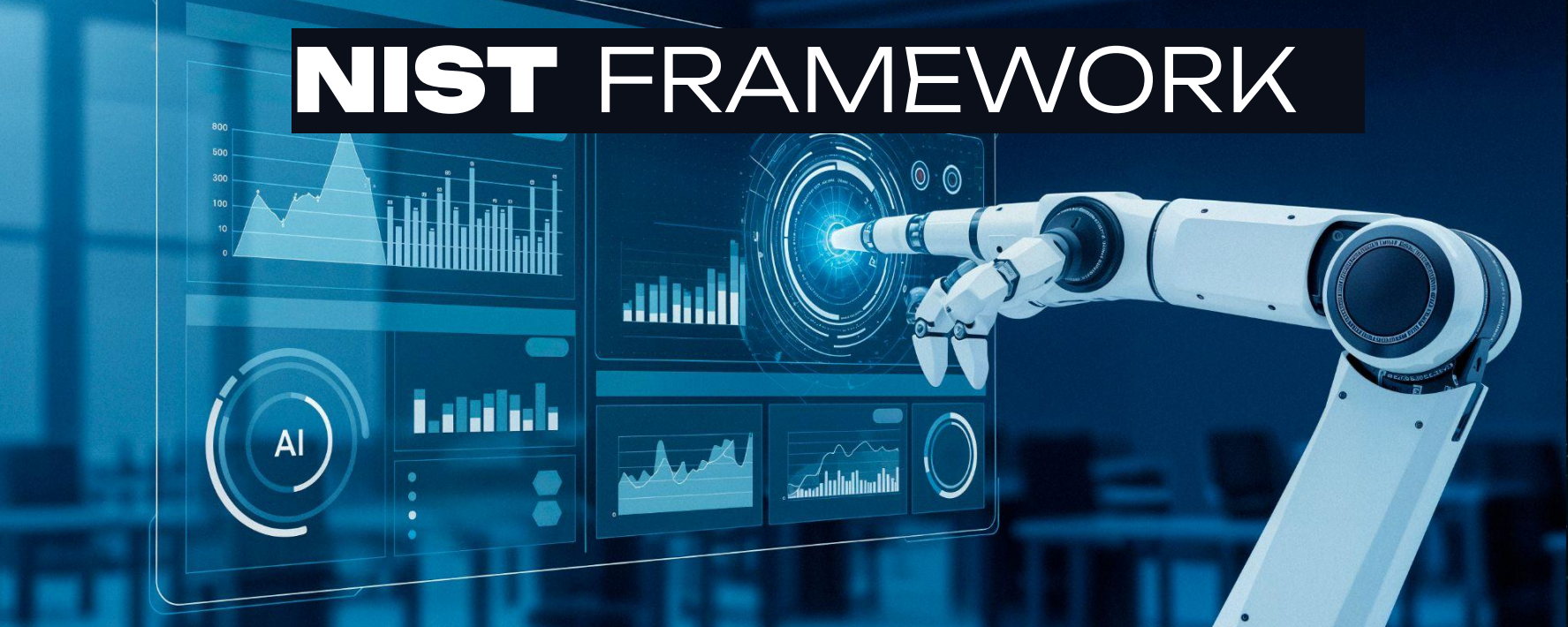
RECOVER

RECOVER

Recovery Steps:

- Restore patient data from the most recent clean backup (daily)
- Rebuild infected servers or systems from a clean, verified image
- Notify affected patients per HIPAA breach-notification rules (within 60 days)
- Conduct a post-incident review and update policies

NIST FRAMEWORK



A flexible, industry-recognized framework that helps organizations improve cybersecurity resilience using.

Identify

Detect

Recover

Protect

Respond